

Foundations of Computing Theory—Homework 5
Due: Wednesday, January 23 2013, 2:00pm

- You only have to explain your answer if this is stated in the question.
 - Before you start on the homework, please read the rules on collaboration and submission in the syllabus.
 - Keep in mind that, as stated in the syllabus, I will stop answering homework questions at 5pm on Tuesday.
1. Give a CFG that generates the language $\{a^i b^k \mid i \neq k\}$. It suffices to give the rules. Your CFG should not be overly complicated. Is your grammar ambiguous? Explain why or why not.
 2. Give a CFG that generates the language $\{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } j \geq i + k\}$. It suffices to give the rules. Your CFG should not be overly complicated.
 3. Give a CFG that generates the language $\{a, b\}^* - \{ww \mid w \in \{a, b\}^*\}$. It suffices to give the rules. Your CFG should not be overly complicated.
 4. Show that the class of context-free languages is not closed under difference. Use either of the following facts:
 - The class of context-free languages is not closed under intersection.
 - The language $\{ww \mid w \in \{a, b\}^*\}$ is not a CFL.
 5. Consider the following grammar G :

$$S \rightarrow aSb \mid aaSb \mid Sb \mid \epsilon.$$

- (a) Give a description of $L(G)$ using set notation.
- (b) Show that G is ambiguous.
- (c) Give an equivalent CFG that is not ambiguous. It suffices to give the rules. Your CFG should not be overly complicated.